

TAKE-HOME ASSIGNMENT — FULL STACK DEVELOPER

Application Workflow Tracker

A complete walkthrough of the system — backend API, frontend interface, workflow rules, and architecture decisions.

Draft

→

Submitted

→

Under Review

→

Approved

/

Rejected

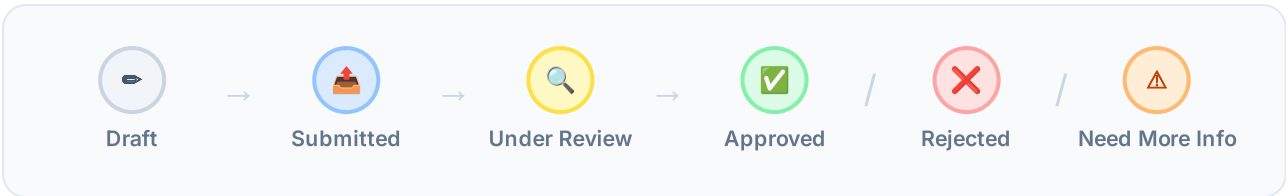
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More Info



System Overview

The Workflow Tracker is a full-stack web application that manages applications through a structured approval lifecycle. Applicants create and submit applications; reviewers evaluate and record decisions.



A **Need More Information** decision sends the application back to the applicant, who edits and resubmits it — re-entering the Under Review stage.

Backend

Django 6 + Django Ninja 1.6 · SQLite · 7 REST endpoints · Auto-generated tracking numbers · Full workflow rule enforcement

Frontend

React 19 + Vite · React Router v6 · Status-based action rendering · Toast notifications · Workflow progress bar · Breadcrumb navigation

Application Types

Recordation · Renewal · Change of Ownership · Change of Name · Discontinuation

Tech Stack

LAYER	TECHNOLOGY	PURPOSE
API Framework	Django Ninja	REST endpoints with automatic Swagger docs and Pydantic validation
Database	SQLite	Zero-config local storage for development
Frontend Build	Vite	Fast dev server with Hot Module Replacement
Routing	React Router v6	Client-side navigation between screens

CORS

django-cors-headers

Allow frontend origin to call the backend API

Application List

The home screen shows all applications with a live status summary bar and a sortable table. Clicking any row opens the detail page.

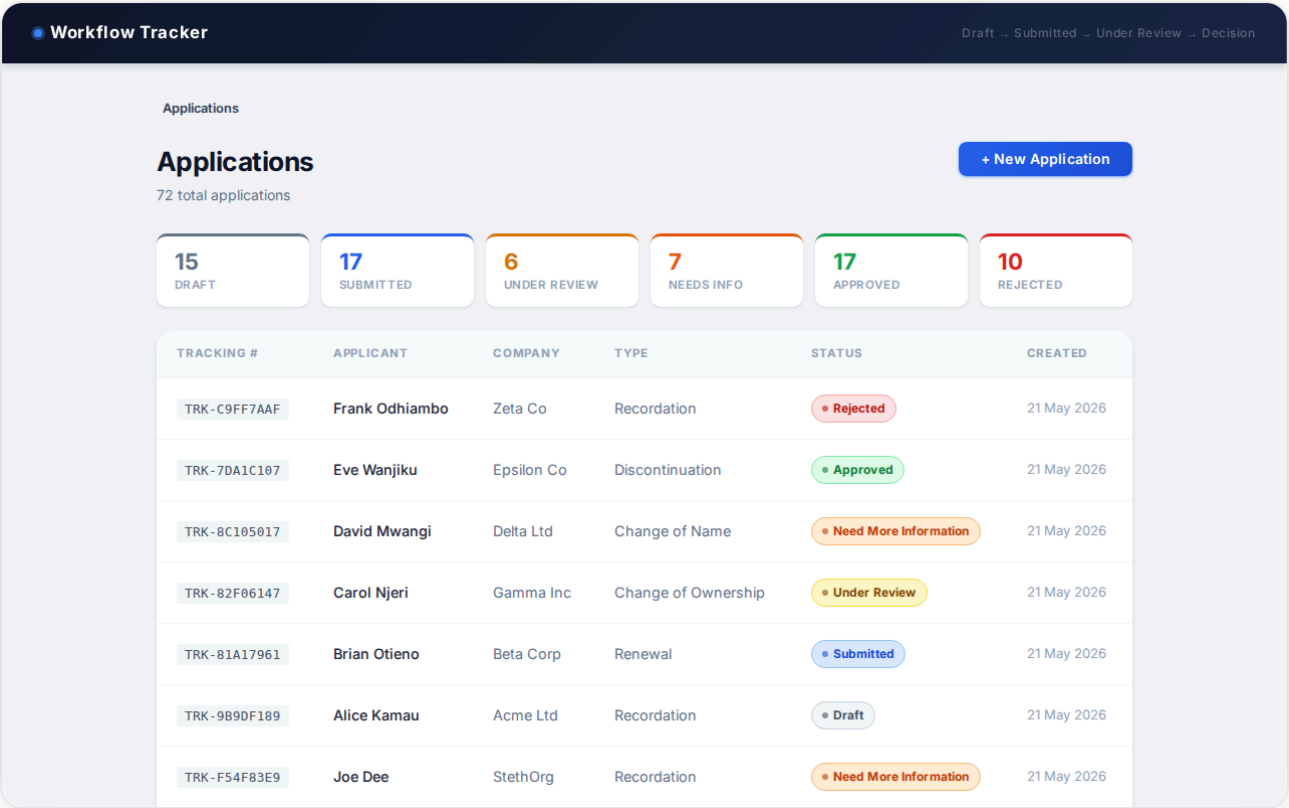


Fig 1 — Application list with stats bar, all six required columns, and colour-coded status badges

The stats bar at the top gives an instant count per status — reviewers can see at a glance how many applications are waiting for action.

Columns displayed

COLUMN	DESCRIPTION
Tracking #	Auto-generated unique identifier (e.g. TRK-9B9DF189)
Applicant	Full name of the applicant
Company	Organisation or company name
Type	Application type (Recordation, Renewal, etc.)
Status	Colour-coded badge — each status has a distinct colour

Status colour coding

- Draft
- Submitted
- Under Review
- Need More Information
- Approved
- Rejected

Creating an Application

Any user can create a new draft application by clicking **+ New Application** on the list screen. The form collects all required fields before saving.

Workflow Tracker

Draft ... Submitted ... Under Review ... Decision

Applications > New Application

NEW APPLICATION

Create Application

Cancel

APPLICANT DETAILS

Full Name *

e.g. Jane Doe

Email Address *

jane@company.com

Company / Organisation *

Company name

Application Type *

— Select type —

APPLICATION DETAILS

Description *

Describe the purpose and details of this application...

Save Draft

Save & Submit

Fig 2 — New application form with two-column layout and required-field indicators

Form fields

FIELD	VALIDATION
Full Name	Required
Email Address	Required, valid email format
Company / Organisation	Required
Application Type	Required — select from five predefined types
Description	Required — free text

Two save options

Save Draft

Workflow Tracker — Application Walkthrough

Save & Submit

github.com/ezko-don/workflow-tracker

Saves the application as a Draft. The applicant can return later to continue editing before submitting.

Saves and immediately submits the application in one step. Status moves to Submitted.

On save, the system auto-generates a unique tracking number (e.g. **TRK-9B9DF189**) and redirects to the detail page with a success toast notification.

Draft Application

The detail screen shows the full application, a visual workflow progress bar indicating the current stage, and context-sensitive action buttons at the bottom.

Workflow Tracker Draft ... Submitted ... Under Review ... Decision

Applications > TRK-9B9DF189

APPLICATION TRK-9B9DF189 Draft Back

Workflow Progress:

- 1 **Draft** (Active)
- 2 Submitted
- 3 Under Review
- 4 Decision

APPLICANT

Name	Alice Kamau
Email	alice@acme.com
Company	Acme Ltd
Type	Recordation

TIMELINE

Created	21 May 2026, 14:49
Last Updated	21 May 2026, 14:49

DESCRIPTION

Annual recordation renewal for trademark portfolio.

Edit Submit Application

Fig 3 — Draft detail page with workflow progress bar, Applicant + Timeline cards, and Edit / Submit actions

Available actions — Draft

Edit

Opens the edit form. All fields can be updated while the application is in Draft status.

Submit Application

Locks the application from further edits and moves it to Submitted, making it visible to reviewers.

Workflow rule: Only Draft applications can be edited or submitted. Attempting either action on a non-Draft application returns a 400 error with a descriptive message.

Workflow progress bar

Every detail page includes a visual step tracker. The active step is highlighted in blue; completed steps turn green; terminal states (Approved / Rejected / NMI) show a coloured icon.

Submitted Application

Once submitted, the application is locked from applicant edits. A reviewer sees a single action — **Start Review** — to take ownership of the application.

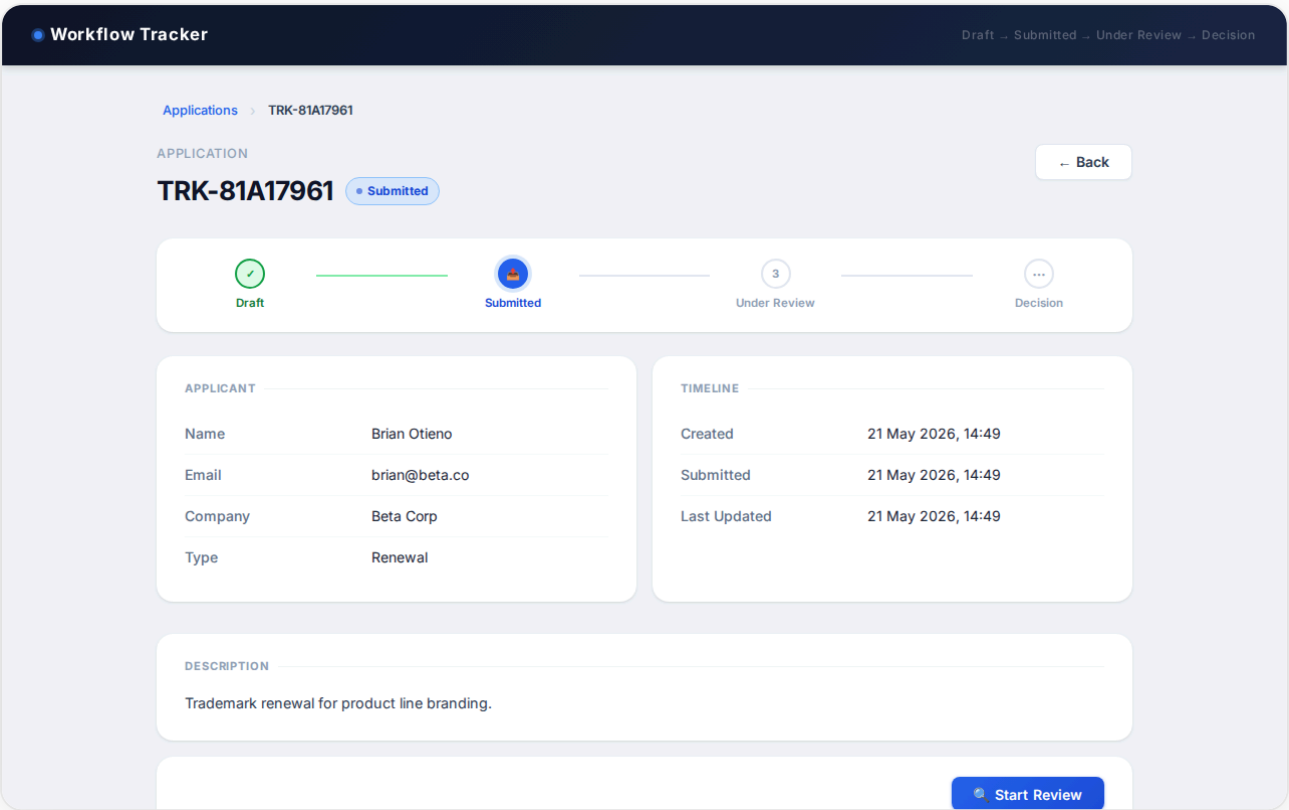


Fig 4 — Submitted application — only the Start Review button is shown; no Edit option

Workflow rule: Only a Submitted application can move to Under Review. If Start Review is clicked on an application in any other status, the API returns a clear error.

What changes on submit

FIELD	BEFORE SUBMIT	AFTER SUBMIT
status	Draft	Submitted
submitted_at	null	Current timestamp
Edit action	Visible	Hidden

Under Review

When a reviewer starts the review, the application moves to Under Review. The reviewer can then record a decision — Approve, Request More Information, or Reject.

The screenshot shows the 'Workflow Tracker' interface. At the top, a dark blue header contains the title 'Workflow Tracker' and a breadcrumb trail: 'Draft → Submitted → Under Review → Decision'. Below the header, the page title 'Applications > TRK-82F06147' is displayed. The main content area is titled 'APPLICATION' and features the application ID 'TRK-82F06147' with a yellow 'Under Review' status badge. A 'Back' button is located to the right. A progress bar below the title shows four stages: 'Draft' (green checkmark), 'Submitted' (green checkmark), 'Under Review' (blue circle with '9'), and 'Decision' (grey circle with three dots). The 'Under Review' stage is currently active. Below the progress bar, there are two side-by-side panels. The left panel, titled 'APPLICANT', contains a table with the following data: Name: Carol Njeri, Email: carol@gamma.co, Company: Gamma Inc, and Type: Change of Ownership. The right panel, titled 'TIMELINE', contains a table with the following data: Created: 21 May 2026, 14:49, Submitted: 21 May 2026, 14:49, and Last Updated: 21 May 2026, 14:49. Below these panels is a 'DESCRIPTION' section with the text: 'Ownership transfer following company acquisition.' At the bottom right of the page, there is a blue button labeled 'Record Decision'.

APPLICANT	
Name	Carol Njeri
Email	carol@gamma.co
Company	Gamma Inc
Type	Change of Ownership

TIMELINE	
Created	21 May 2026, 14:49
Submitted	21 May 2026, 14:49
Last Updated	21 May 2026, 14:49

DESCRIPTION

Ownership transfer following company acquisition.

Record Decision

Fig 5 — Under Review application with Record Decision button

Reviewer Decision Modal

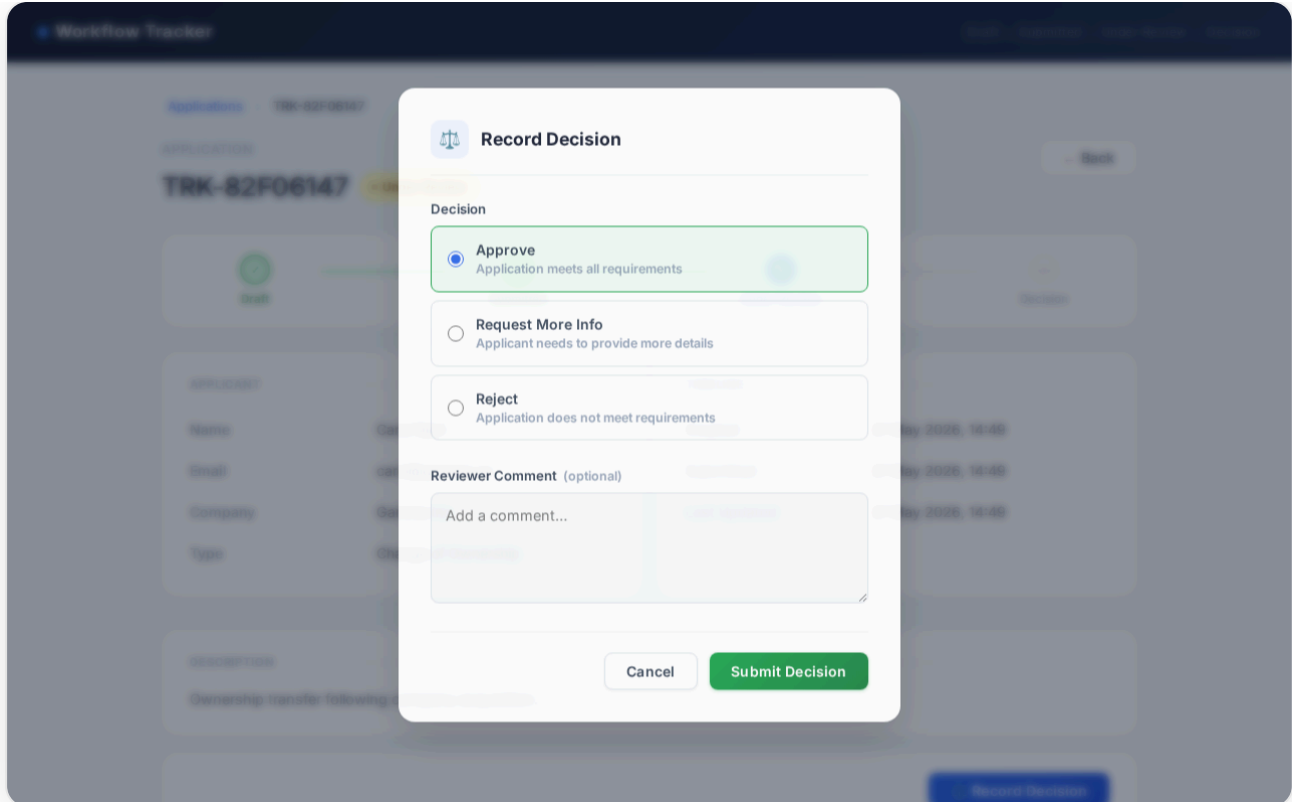


Fig 6 — Decision modal with radio card options and reviewer comment field

The modal presents three clearly described options. Selecting **Request More Info** or **Reject** makes the comment field required. The submit button colour changes to match the selected decision (green / amber / red).

Workflow rule: A reviewer comment is mandatory for Need More Information and Rejected decisions. The Submit Decision button remains disabled until a comment is entered.

Need More Information

When a reviewer requests more information, the application is returned to the applicant. The reviewer's comment is prominently displayed so the applicant knows exactly what to address.

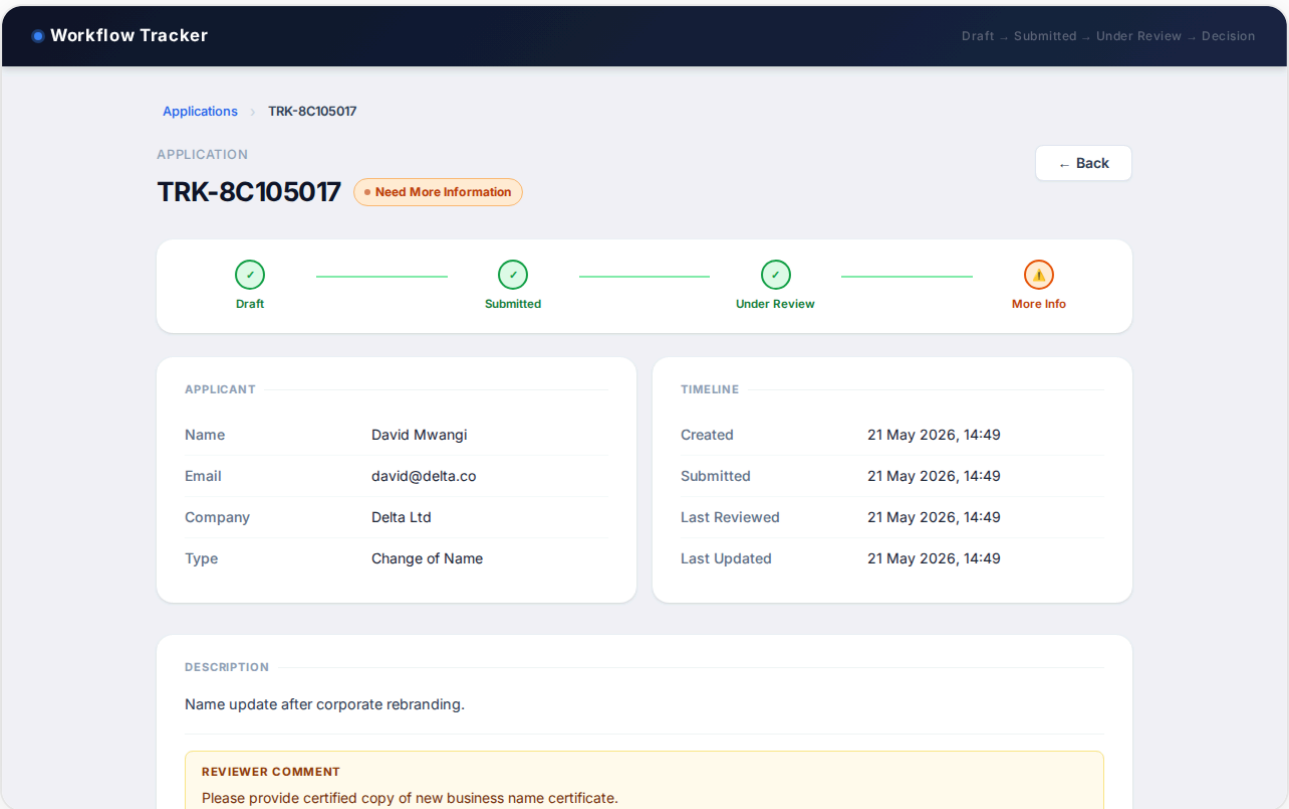
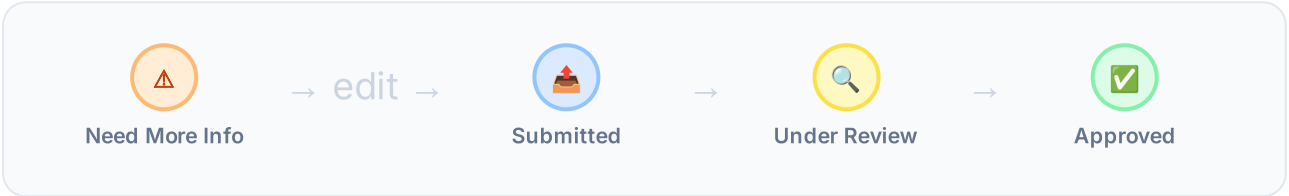


Fig 7 — NMI application showing the reviewer's comment in an amber highlight box and the Edit & Resubmit button

The Edit & Resubmit flow: clicking the button opens the edit form pre-populated with existing data. Saving resubmits the application directly, moving it back to Submitted status.

NMI resubmit loop



This loop can repeat as many times as needed until the reviewer is satisfied and approves or rejects the application.

Approved & Rejected

Approved and Rejected are terminal states. No further actions are available — the action bar shows a closed message and all edit/submit routes are blocked at the API level.

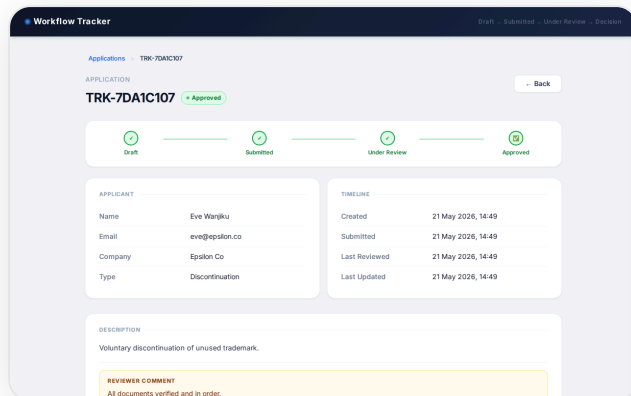


Fig 8 — Approved application — green terminal state, no action buttons

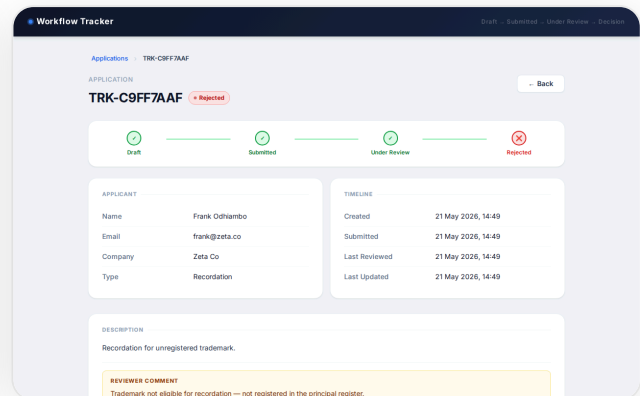


Fig 9 — Rejected application with reviewer comment — no action buttons

Workflow rule: Approved and Rejected applications cannot be edited. Any PUT request to the update endpoint returns 400 — "Applications with status 'Approved' cannot be edited."

Toast Notifications & Breadcrumbs

Every user action produces an immediate toast notification — green for success, red for errors. Breadcrumb navigation appears on all inner pages so users always know where they are.

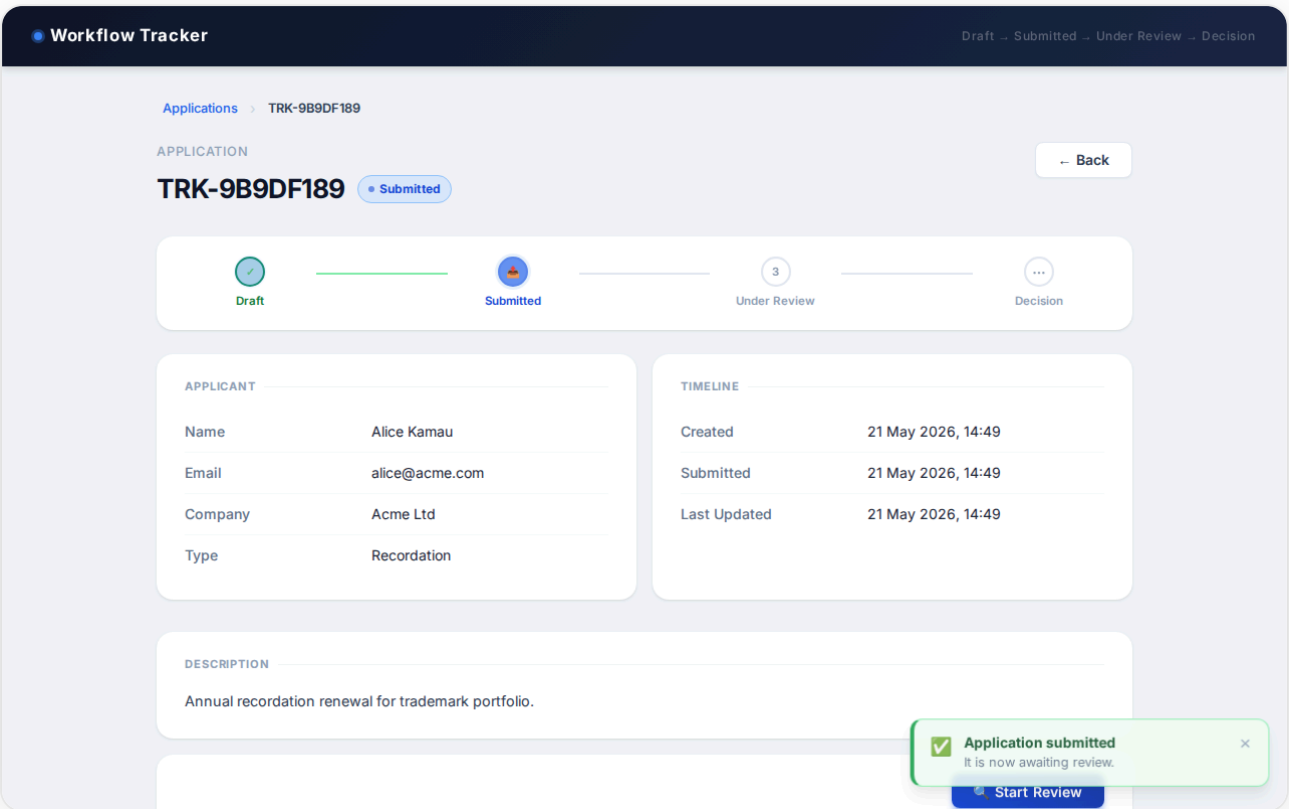


Fig 10 — Success toast after submitting an application, showing the workflow bar updating to Submitted in real time

Toast triggers

ACTION	TOAST
Create draft	✔ "Draft created" + tracking number
Save draft	✔ "Draft saved"
Submit	✔ "Application submitted — awaiting review"
Start review	✔ "Review started"
Approve	✔ "Approved"
Need More Info	⚠ "Returned for more info" + reviewer comment
Reject	✖ "Rejected"

Any API error

 Red error toast with the error message

Breadcrumb paths

SCREEN	BREADCRUMB
List	Applications
Detail	Applications > TRK-XXXXXXXX
Create form	Applications > New Application
Edit form	Applications > TRK-XXXXXXXX > Edit
Edit & Resubmit	Applications > TRK-XXXXXXXX > Edit & Resubmit

REST API Reference

The backend is a Django Ninja REST API. Interactive Swagger documentation is available at **<http://localhost:8000/api/docs>** when the server is running.

METHOD	ENDPOINT	DESCRIPTION
POST	<code>/api/applications/</code>	Create a new draft application
GET	<code>/api/applications/</code>	List all applications
GET	<code>/api/applications/{id}</code>	View application details
PUT	<code>/api/applications/{id}</code>	Update a draft application
POST	<code>/api/applications/{id}/submit</code>	Submit application for review
POST	<code>/api/applications/{id}/review</code>	Move to Under Review
POST	<code>/api/applications/{id}/decide</code>	Record reviewer decision

Workflow rules enforced at API level

RULE	HTTP RESPONSE WHEN VIOLATED
Only Draft can be edited	400 — "Applications with status '...' cannot be edited."
Only Draft / NMI can be submitted	400 — "Only Draft or Need More Information applications can be submitted."
Only Submitted → Under Review	400 — "Only Submitted applications can be moved to Under Review."
Only Under Review gets a decision	400 — "Only Under Review applications can receive a decision."
Comment required for NMI / Rejected	400 — "A reviewer comment is required for this decision."

Project structure

FILE	PURPOSE
<code>applications/models.py</code>	Application model, ApplicationType and Status choices

applications/schemas.py	Pydantic request/response schemas (Django Ninja)
applications/api.py	All 7 endpoints + workflow rule enforcement
applications/tests.py	Model unit tests (default status, tracking number, transitions)
frontend/src/api/applications.js	All fetch calls in one place
frontend/src/context/ToastContext.jsx	Global toast notification state
frontend/src/components/WorkflowProgress.jsx	Visual step tracker on detail pages

